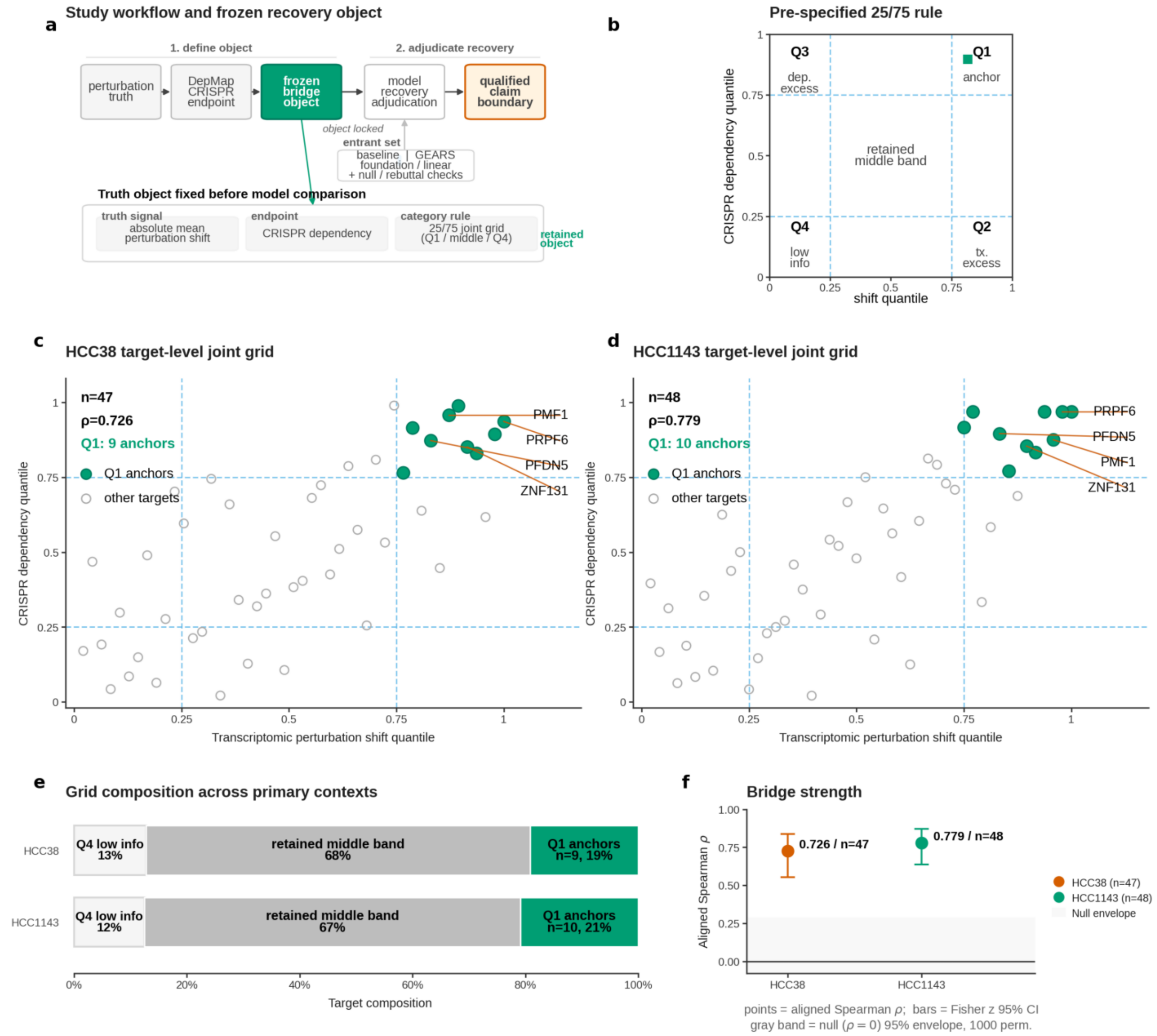


Fig. 1 | Truth object definition. a, Workflow defining the frozen phenotype-aligned recovery object before model scoring by linking absolute mean perturbation shift to CRISPR DepMap dependency. b, Pre-specified 25/75 joint-percentile rule separating Q1 anchors, Q2 transcriptomic excess, Q3 dependency excess, Q4 low-information targets and the retained middle band. c,d, HCC38 and HCC1143 target-level rank-percentile grids; panels report n, aligned Spearman rho and Q1 anchor count. e, Category composition across the two primary contexts, including zero-count Q2 and Q3 categories. f, Bridge strength summary showing observed aligned Spearman rho with Fisher z 95% confidence intervals and a 1,000-permutation null envelope.

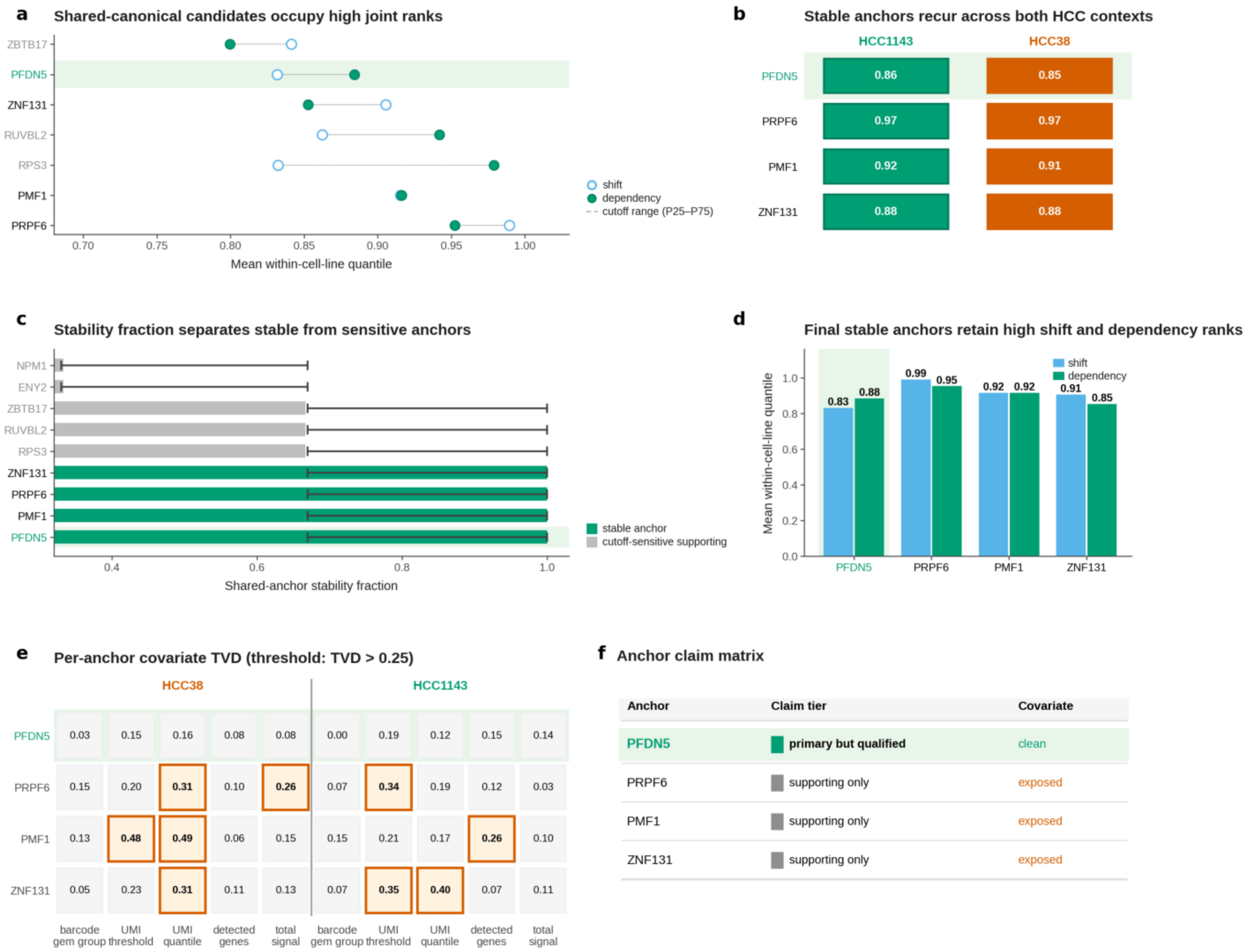


Fig. 2 | Anchor tiering. a, Shared-canonical anchor ranking by paired shift and dependency quantile means across HCC38 and HCC1143. b, Recurrence matrix for the four final stable anchors, with Q1 status and mean joint quantiles shown per context. c, Stability fraction separates stable anchors from cutoff-sensitive supporting objects without using covariate information. d, Stable anchors retain high shift and dependency ranks across both HCC contexts. e, Per-anchor covariate Total Variation Distance (TVD) audit across five covariate axes and two HCC contexts; TVD > 0.25 marks imbalance. f, Claim-tier matrix: PFDN5 is primary but qualified, whereas PMF1, PRPF6 and ZNF131 remain supporting because of covariate exposure.

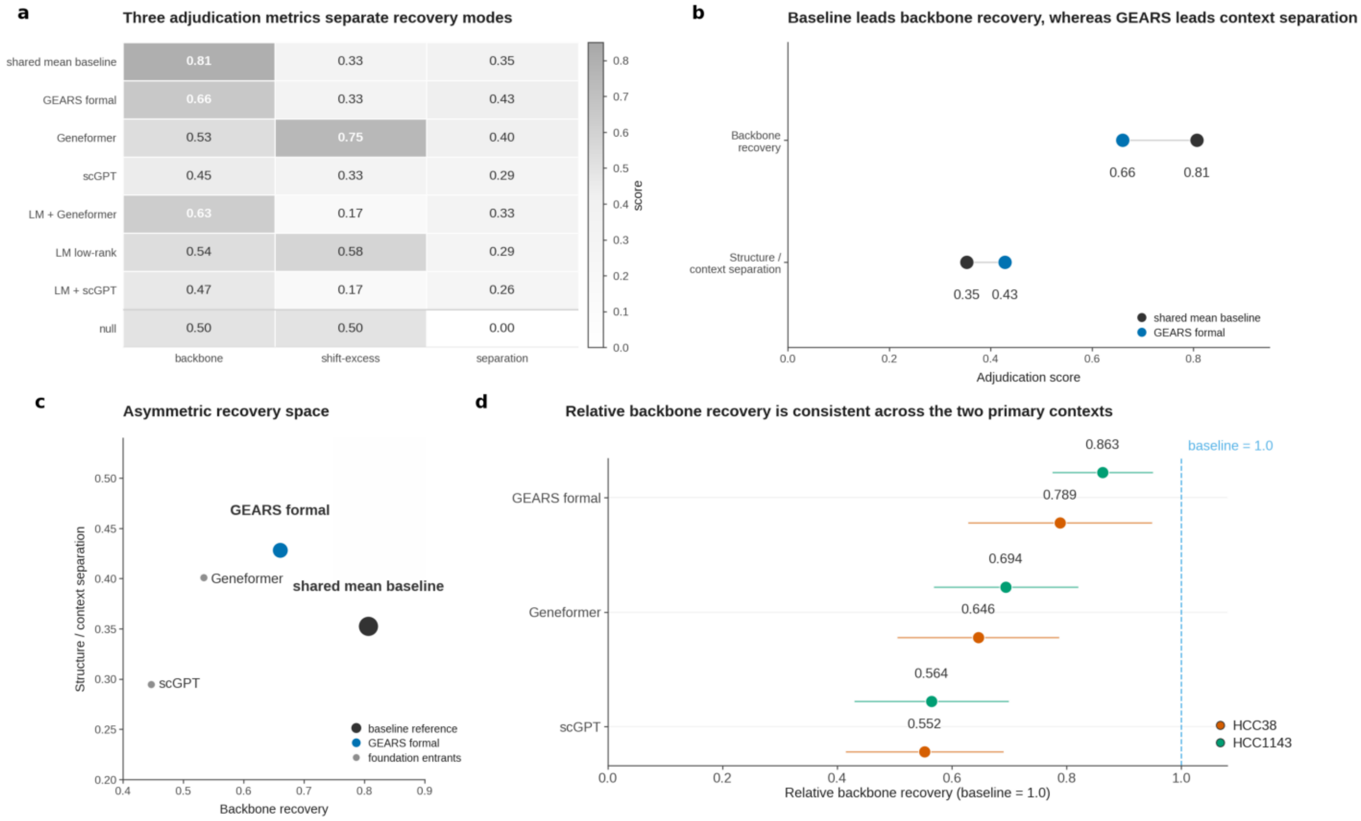
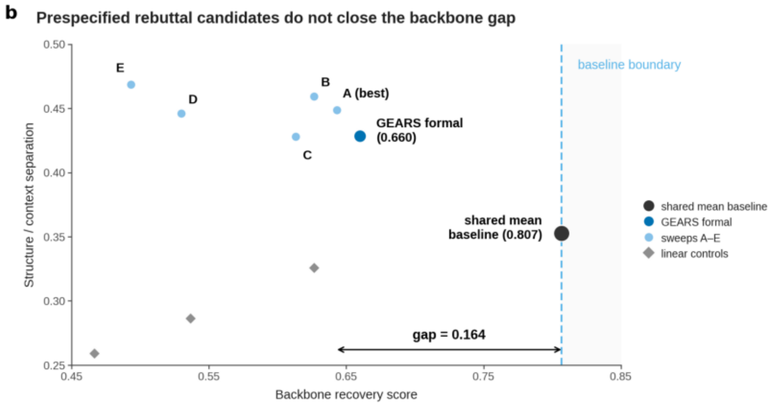


Fig. 3 | Model adjudication. a, Three pre-specified metrics evaluate backbone recovery, shift-excess identification and structure-versus-context separation across the shared-mean baseline, GEARS, foundation-model entrants, linear controls and a null reference. b, Paired summary showing that the shared-mean baseline leads backbone recovery, whereas GEARS leads structure-versus-context separation. c, Backbone recovery versus structure-versus-context separation places entrants in an asymmetric recovery space; no entrant occupies the illustrative upper-right region, so GEARS is retained as an architecture-level diagnosis rather than an overall primary winner.

a Pre-specified finite-budget GEARS candidates

Candidate	Epoch	LR	WD	Role
GEARS formal	30	1×10^{-3}	1×10^{-6}	reference recipe
A (best)	30	2×10^{-3}	1×10^{-6}	best sweep candidate
B	20	1×10^{-3}	1×10^{-6}	sweep candidate
C	30	1×10^{-3}	1×10^{-5}	sweep candidate
D	30	5×10^{-4}	1×10^{-6}	sweep candidate
E	40	1×10^{-3}	1×10^{-6}	sweep candidate



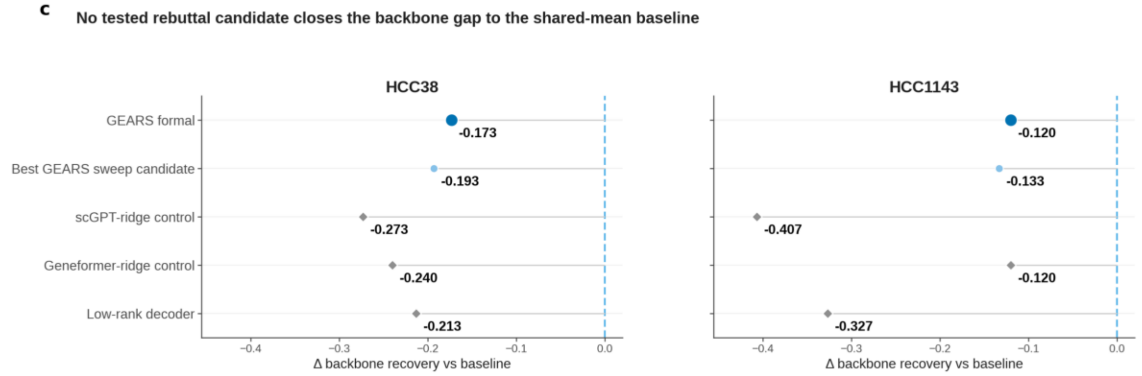


Fig. 4 | Finite-budget rebuttal tests. a, Six pre-specified GEARS recipes define the finite local rebuttal neighborhood under the unchanged truth object and scoring system. b, Rebuttal recovery space compares the shared-mean baseline, formal GEARS, five GEARS sweep candidates and three embedding-based linear controls on backbone recovery and structure/context separation. c, Per-context residual backbone gaps show that all tested candidates remain below the shared-mean baseline in HCC38 and HCC1143; the stop rule therefore does not promote GEARS to the primary winner.

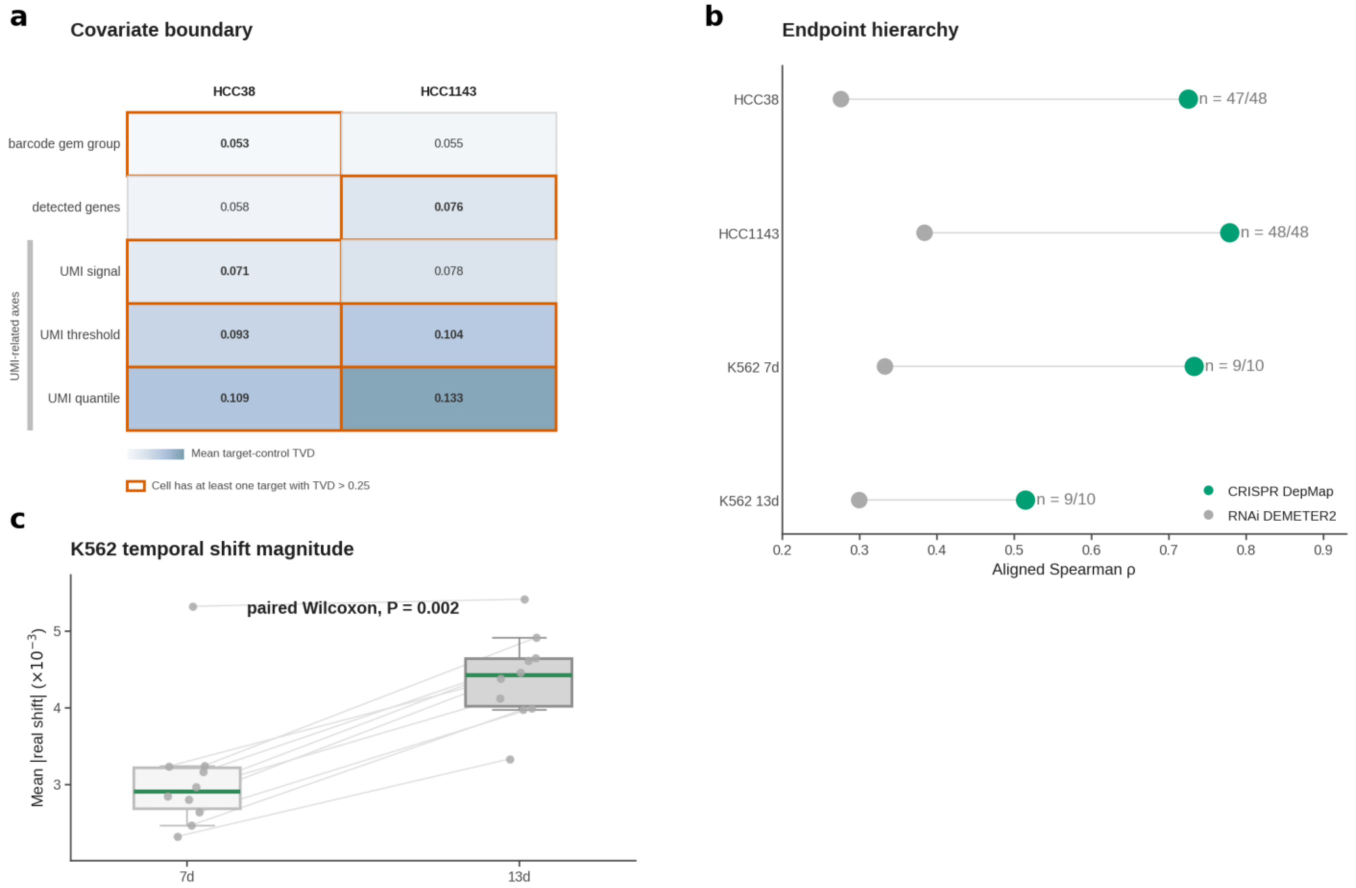
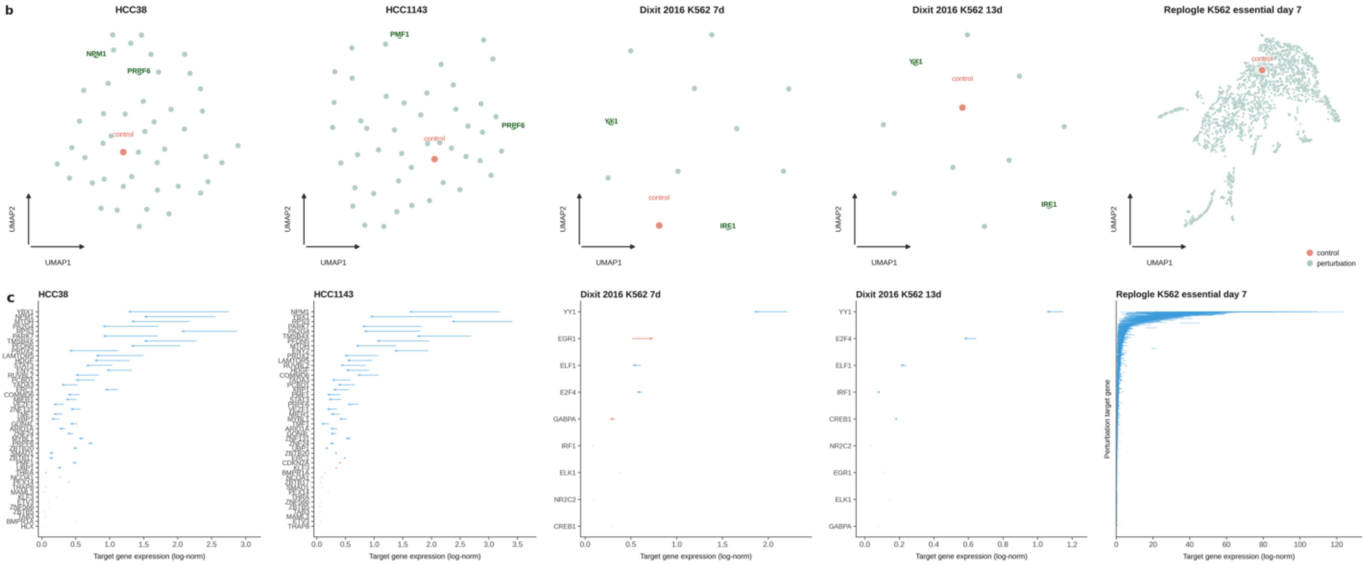


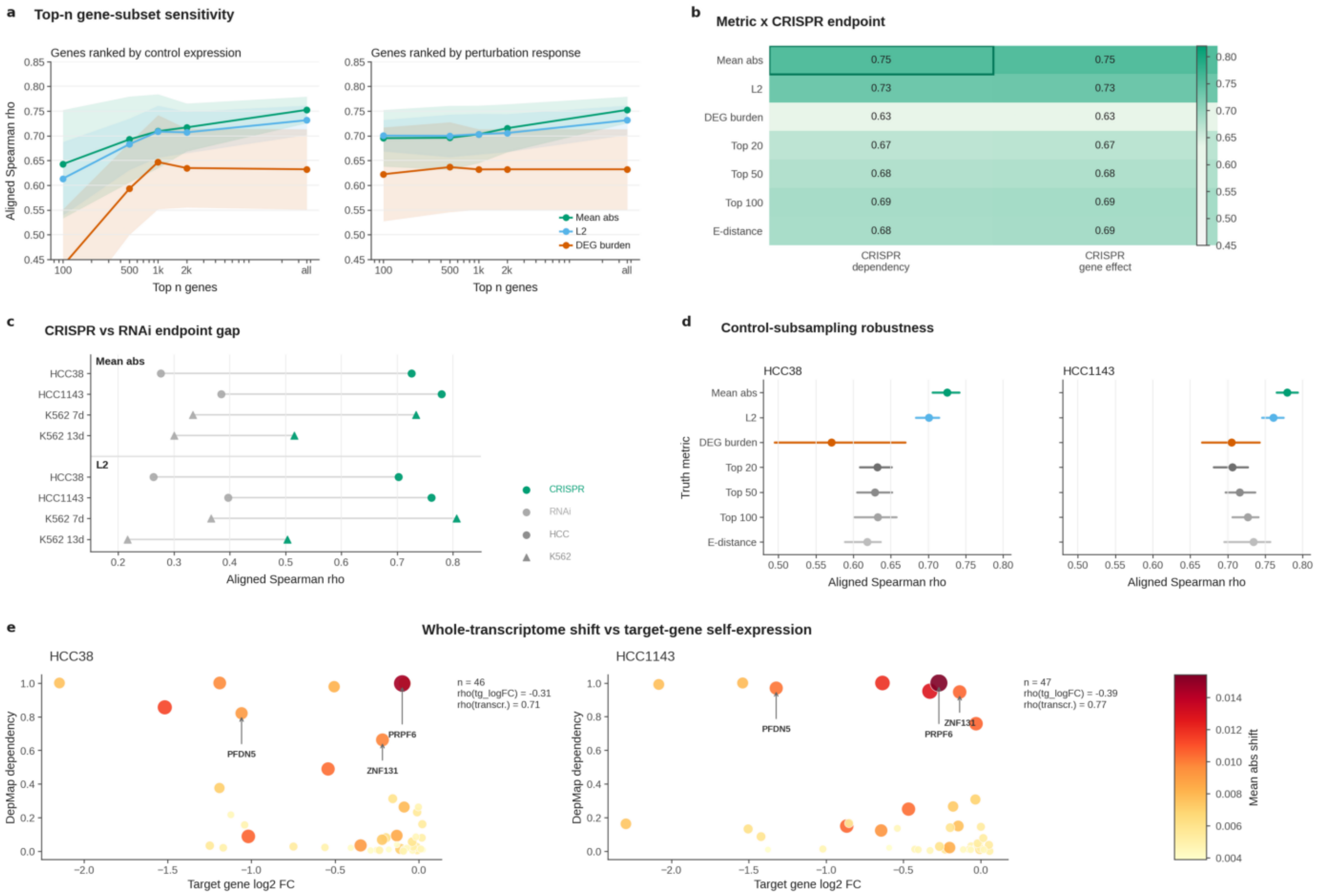
Fig. 5 | Boundary governance. a, Covariate boundary in HCC38 and HCC1143, showing mean target-control TVD across five covariate stratifications and marking targets exceeding TVD > 0.25. b, Endpoint hierarchy across HCC38, HCC1143, K562 7d and K562 13d compares CRISPR DepMap dependency with RNAi DEMETER2; CRISPR is higher in all four contexts. c, K562 temporal stratification shows larger perturbation magnitude at 13d but stronger dependency-aligned rank structure at 7d. d, A0/A1/B tiering retains K562 as architecture-form and bounded bridge-form support, not content-level replication.

a

Dataset	Size	Benchmark use
Perturbation datasets		
HCC38	36,601 genes x 14,175 cells	47 targets | 1,666 controls
HCC1143	36,601 genes x 11,405 cells	47 targets | 1,325 controls
Dixit 2016 K562 7d	20,824 genes x 28,034 cells	10 targets | 5,381 controls
Dixit 2016 K562 13d	20,824 genes x 15,849 cells	10 targets | 3,491 controls
Replogle K562 essential day 7	8,563 genes x 310,385 cells	2057 targets | 10,691 controls
Endpoint datasets		
DepMap CRISPR dependency (primary endpoint)	1,186 cell lines x 18,435 genes	HCC38 47 / HCC1143 48 / K562 7d 10 / K562 13d 10 / Replogle K562 1,882
DEMETER2 RNAi (sensitivity endpoint)	701 mapped cell lines x 17,309 genes	HCC38 48 / HCC1143 48 / K562 7d 9 / K562 13d 9 / Replogle K562 1,882

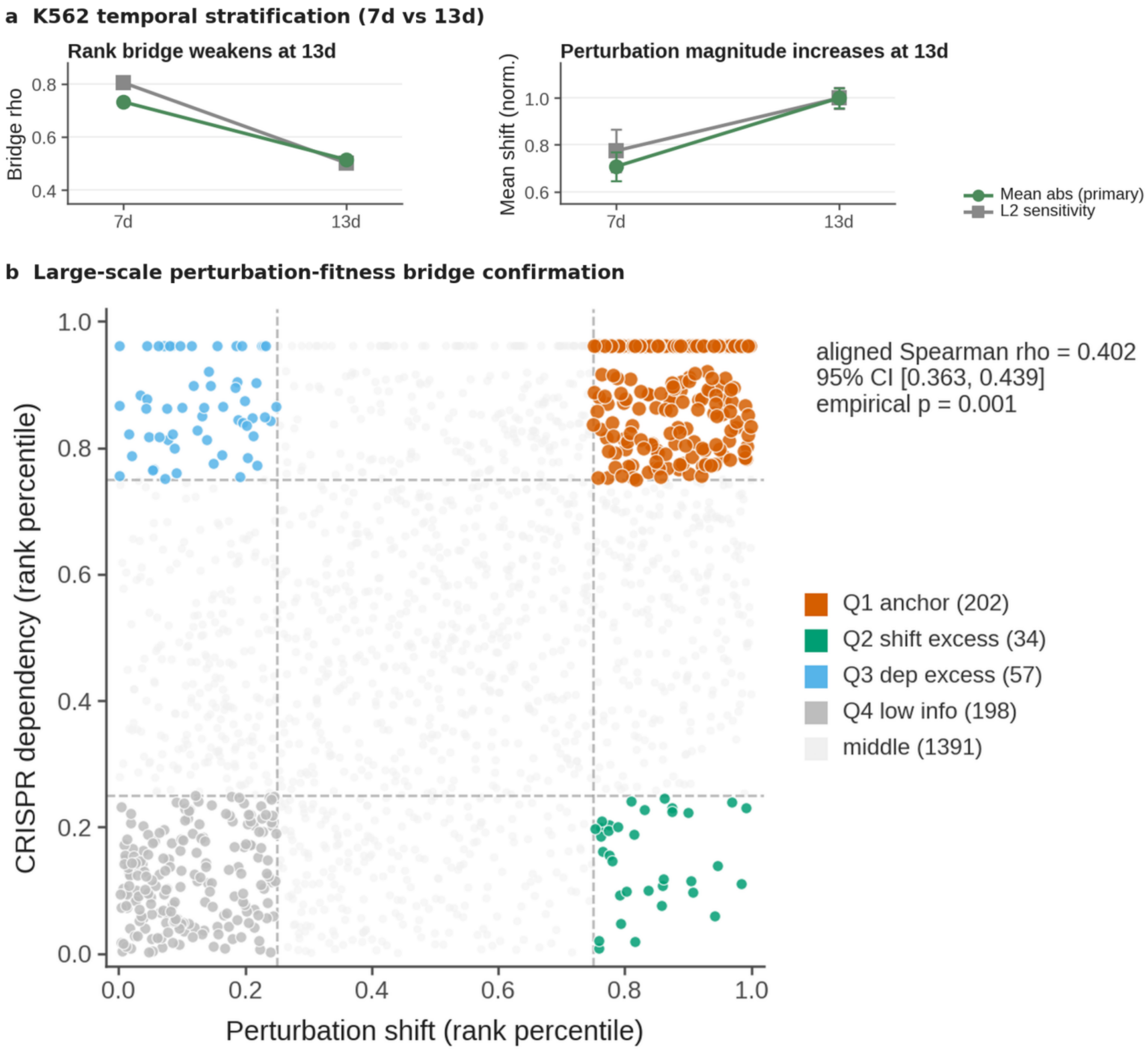


Extended Data Fig. 1 | Dataset familiarization.
a, Dataset overview for five perturbation-expression contexts and two endpoint resources, summarizing cell-line identity, size and benchmark use. b, UMAPs of perturbation-level mean profiles for HCC38, HCC1143, Dixit K562 7d, Dixit K562 13d and Replogle K562 essential day 7; the matched control aggregate is marked. c, Target-gene expression changes for the same five contexts, with arrows from control expression to post-perturbation expression.

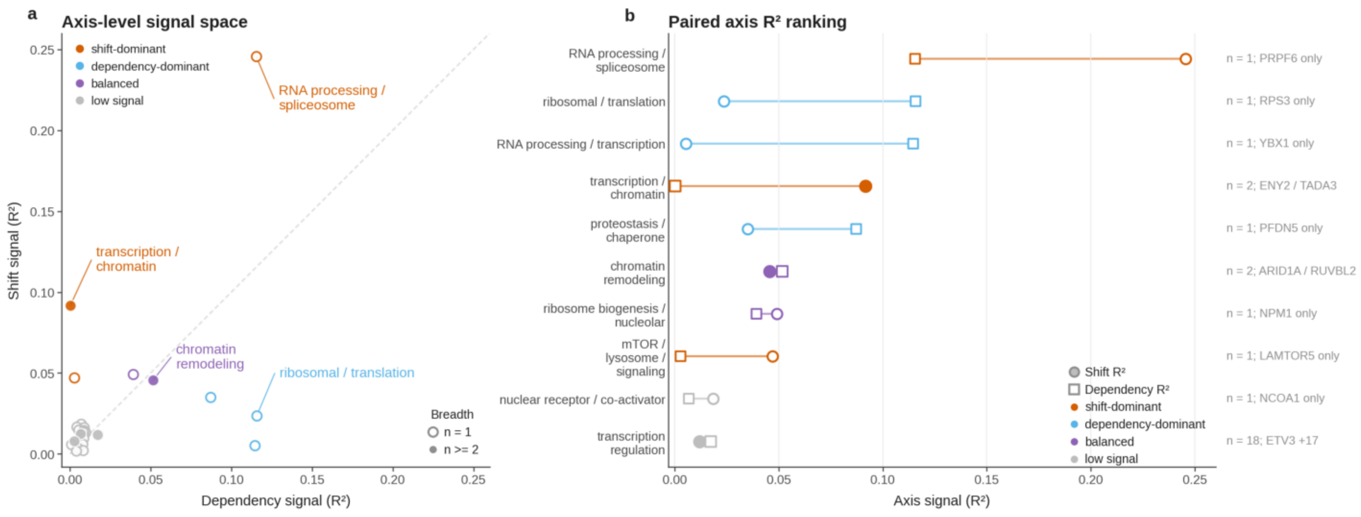


Extended Data Fig. 2 | Metric robustness.

a, Top-n gene-subset sensitivity recomputes aligned Spearman correlation after ranking genes by control expression or perturbation-response magnitude. b, Metric and CRISPR endpoint heat map marks the retained primary bridge metric and endpoint. c, Endpoint sensitivity compares CRISPR dependency with RNAi DEMETER2 across primary HCC and supplementary K562 contexts. d, Control-subsampling robustness shows mean and 2.5th-97.5th percentile ranges across repeated control-cell subsamples. e, Whole-transcriptome shift, rather than target-gene self-expression, carries the fitness bridge.

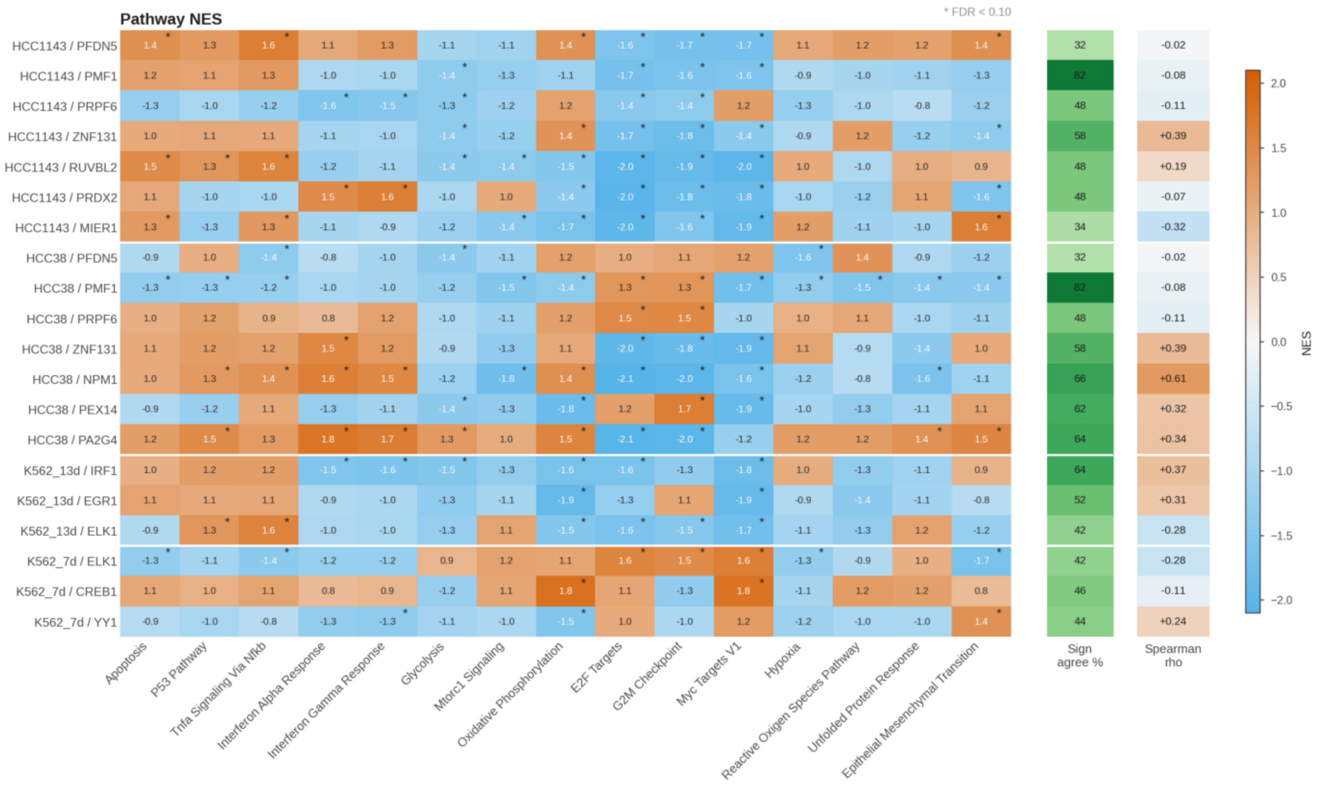


Extended Data Fig. 3 | K562 and Replogle bridge support.
a, K562 temporal bridge-magnitude dissociation shows aligned Spearman correlation with CRISPR DepMap dependency and mean perturbation-shift magnitude at 7d and 13d; 13d has larger magnitude but weaker rank alignment. b, Replogle K562 essential day 7 bridge test plots 1,882 matched CRISPRi targets as rank percentiles for perturbation shift and dependency, with 25/75 percentile cutoffs, Q1-Q4 category counts and aligned Spearman rho = 0.402.



Extended Data Fig. 4 | Axis-level signal space.
a, Axis-level signal space compares dependency signal with transcriptomic shift signal across annotated axes; color denotes descriptive signal profile rather than claim tier. b, Paired axis R-squared ranking displays axes with the largest signal in either dimension on a common scale, supporting an audit-style view of partial axis-level structure.

a **Exploratory pathway-level summaries of target-control perturbation responses**



Extended Data Fig. 5 | Pathway-response polarity.

a, Pathway normalized enrichment score heat map across pre-specified display targets in HCC38, HCC1143, K562 7d and K562 13d. Rows show selected anchors and high-variance response examples; columns show the Hallmark response panel; asterisks mark FDR < 0.10. Right-hand summaries report same-target partner-context pathway Spearman correlation and sign agreement, treating pathway polarity as exploratory context rather than a benchmark-defining endpoint.